Negar Rahimi

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Ph.D. candidate at the University of Colorado Boulder specializing in neurophysiology, physiological signal processing, digital health and wearable sensing technologies, with experience in clinical trials and strong expertise in machine learning and deep learning for biomedical data analysis.

EDUCATION

- **Ph.D. | Integrative Physiology (GPA: 3.948)** 2023-Present

  University of Colorado Boulder, CO, USA

  Thesis: Balance control during quiet standing.

- **M.Sc. | Biomedical Engineering (Bioelectric) (GPA: 3.31)** 2019-2023

  Amirkabir University of Technology

  Thesis: Modeling the impact of visual deprivation during walking on the treadmill.

- **B.Sc. | Biomedical Engineering (Bioelectric) (GPA: 3.12)** 2015-2019

  Amirkabir University of Technology

  Thesis: Anti-saccade detector design based on recorded videos by webcam.

TECHNICAL SKILLS

- **Signal Collecting & Processing:** EMG | Balance Tracking Systems (BTS) | CoP | Force Sensor | Wearable Sensors | IMU
- **Programming Languages:** MATLAB | Python | MATLAB Simulink Modelling | C/C++
- **Advanced Data Analysis:** Machine Learning | Deep Learning | Statistical Analysis | LLMs
- **Clinical Trials & Research:** Experiment Design | Data Collection
- **Neurorehabilitation & Motor Control:** TENS | NMES | Force | Balance
- **Frameworks & Libraries:** TensorFlow | PyTorch | Scikit-learn
- **Biomedical Device Design & Software**: ARM | PCB | Spike2 | OTbio | Altium Designer | STM32

EXPERIENCE

- **Research Assistant | Neurophysiology of Movement Lab** 2023-Present

  University of Colorado Boulder, CO, USA.

  - Conducted clinical trials investigating the effects of transcutaneous electrical nerve stimulation (TENS) on neuromuscular function in individuals with Multiple Sclerosis (MS), using multimodal physiological and biomechanical measurements including IMU, force sensors, and CoP analysis.
  - Applied deep learning and machine learning techniques to analyze EMG and balance data to identify patterns of neuromuscular activity.

- Collaborated with multidisciplinary teams on experimental design, machine learning and deep learning implementation, and multimodal dataset development integrating EMG, motor unit activity, IMU, force, and CoP measurements.
- Managed the full lifecycle of research protocols, including experimental design, participant data collection, statistical analysis, and dissemination of results through scientific reports and presentations.

• **Research & Development Engineer |** Salamat Andishan MomtazMad Co. 2021-2022

- Developed synergy- and TENS-based neuromodulation systems for pain reduction and supported the manufacturing and validation of biomedical stimulation devices to ensure functionality and reliability.
- Developed embedded hardware and software, including circuit design, microcontroller programming, and system integration for neuromodulation devices.

• **Research Assistant | Cybernetics, Neurocognition, & Movement Lab.** 2019-2023

Amirkabir University of Technology

- Developed computational model to study the effects of visual deprivation on gait stability during treadmill walking. Investigated changes in lower-limb coordination and balance control using biomechanical motion capture data.

• **Teaching Assistant / Laboratory Instructor**

University of Colorado Boulder, CO, USA.

- Medical Terminology (Spring 2026) | Endocrinology (Spring 2026) | Human Anatomy (Fall 2025)

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- Instrumentation & Medical Measurement Lab (Fall & Spring 2022) | Electronics Lab I (Fall 2021) | Electrical Circuit Lab I (Spring 2021) | Electrical Circuits I (Spring 2021 & Fall 2022)

AWARDS

• Shurl & Kay Curci Foundation PhD **Scholarship.** 2023-2025

University of Colorado Boulder CO, USA.

• CU Boulder, Department of Integrative Physiology. Graduate Student Travel Award. 2025

University of Colorado Boulder CO, USA.

• Shurl & Kay Curci Foundation PhD Scholarship. Travel Award. 2025

University of Colorado Boulder CO, USA.

PUBLICATIONS

Peer-reviewed Publications

• **N. Rahimi,** V. Hatzitaki, A. Kamankesh, A. Gavriilidou, and RM. Enoka. "Machine-learning classification of postural sway in young adults during colored noisy vestibular stimulation" Journal of Electromyography and Kinesiology (2026); DOI: https://doi.org/10.1016/j.jelekin.2026.103125

- **N. Rahimi**, A. Kamankesh, I. G. Amiridis, S. Daneshgar, Ch. Sahinis, V. Hatzitaki, and R. M. Enoka, "Distinguishing Among Standing Postures with Machine Learning-based Classification Algorithms", Experimental Brain Research.243(1) (2025) 1-13.DOI: https://doi.org/10.1007/s00221-024-06959-9
- Kamankesh, A., **Rahimi, N.**, Amiridis, I.G., Sahinis, C., Hatzitaki, V. and Enoka, R.M., 2024. Distinguishing the Activity of Flexor Digitorum Brevis and Soleus Across Standing Postures with Deep Learning Models. Gait & Posture. DOI: https://doi.org/10.1016/j.gaitpost.2024.12.014
- O. Shoja, M. Shojaei, H. Hassanlouei, F. Towhidkhah, M. Amiri, H. Boroomand, **N. Rahimi**, and L. Zhang. "Lack of visual information alters lower limb motor coordination to control center of mass trajectory during walking" Journal of Biomechanics 155 (2023): 111650. DOI: https://doi.org/10.1016/j.jbiomech.2023.111650

Conference Abstracts

- **N. Rahimi,** M. Henry, R. Al-Jaafari, B. Kott, N. Toninelli, I. Cleveland, K. Weiser, E. Riewe, H. Riewe, and R. M. Enoka. "Influence of treatment with TENS on postural control in Individual with multiple sclerosis". 16th Rocky Mountain American Society of Biomechanics Conference 2026.
- A. Kamankesh, S. Daneshgar, **N. Rahimi**, and R. M. Enoka. "MMNET: an AI-based framework for extracting motor-unit modes from EMG signals". 16th Rocky Mountain American Society of Biomechanics Conference 2026.
- **N. Rahimi,** V. Hatzitaki, A. Kamankesh, A. Gavriilidou, and RM. Enoka. "Classifying the center of pressure trajectory under colored, noisy galvanic vestibular stimulations using machine learning techniques". Society for Neuroscience (SfN) Conference 2025.
- **N. Rahimi**, A. Kamankesh, I. G. Amiridis, S. Daneshgar, Ch. Sahinis, V. Hatzitaki, and R. M. Enoka, "Machine learning-based classification of standing postures". 15th Rocky Mountain American Society of Biomechanics Conference 2025.
- A. Kamankesh, A. Bagheri, **N. Rahimi**, M. Ahmadi Pajouh. "Classification of Finger Movements from EMG Signals Using Deep Neural Networks". 14th Rocky Mountain American Society of Biomechanics Conference 2024.

SERVICE

- Peer Reviewer, Discover Applied Sciences (Springer Nature), (Reviewed 1 manuscript)
- Peer Reviewer, Scientific Reports (Springer Nature), (Reviewed 3 manuscripts)
- Peer Reviewer, Medicine & Science in Sports & Exercise (MSSE), (Reviewed 5 manuscripts)
- Peer Reviewer, Journal of King Saud University Computer & Information Sciences (Springer Nature), (Reviewed 1 manuscript)
- Peer Reviewer, Journal of NeuroEngineering & Rehabilitation (Springer Nature), (Reviewed 2 manuscripts)

LANGUAGES

- **English:** Professional proficiency • **Persian:** Native • **Spanish & Arabic:** Elementary proficiency

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VOLUNTEER EXPERIENCE

• Served as a mentor for the summer 2026 STEM Research Experience Program at the CU Boulder.

• Served as a mentor for the 2025–2026 Graduate Peer Mentoring Program at the CU Boulder.

• Volunteer BCD Science Fair Judge for Boulder Country Day School's Science Fair (2024).

MEDIA COVERAGE

• My research has been highlighted in media such as the Office of Advancement, CU Boulder.